

LIGHTING SYSTEM

Flexible open source automated quality control for industry 5.0 using artificial intelligence and computer vision

1. Label images with Date Set.

This guide explains how to set up a Label Studio environment and configure a YOLO project.

1.1. Prerequisites

Before installing the tool, the following must be installed on the system:

- Docker
- (Optional) Docker Compose

To verify Docker installation, run in terminal:

```
docker --version
```

If installed, the version will be displayed.

1.2. Installing Label Studio via Docker

The easiest way to run Label Studio is to use the official Docker image.

1.3. Download Docker image

Docker images are usually downloaded from the official docker hub, the following command will download the Label Studio image from there:

```
docker pull heartexlabs/label-studio:latest
```

1.4. Run the container

The command we need to use to run the container will be:

```
docker run -it -p 8080:8080 -v  
$(pwd)/label-studio-data:/label-studio/data  
heartexlabs/label-studio:latest
```

The explanation of the parameters used would be as follows:

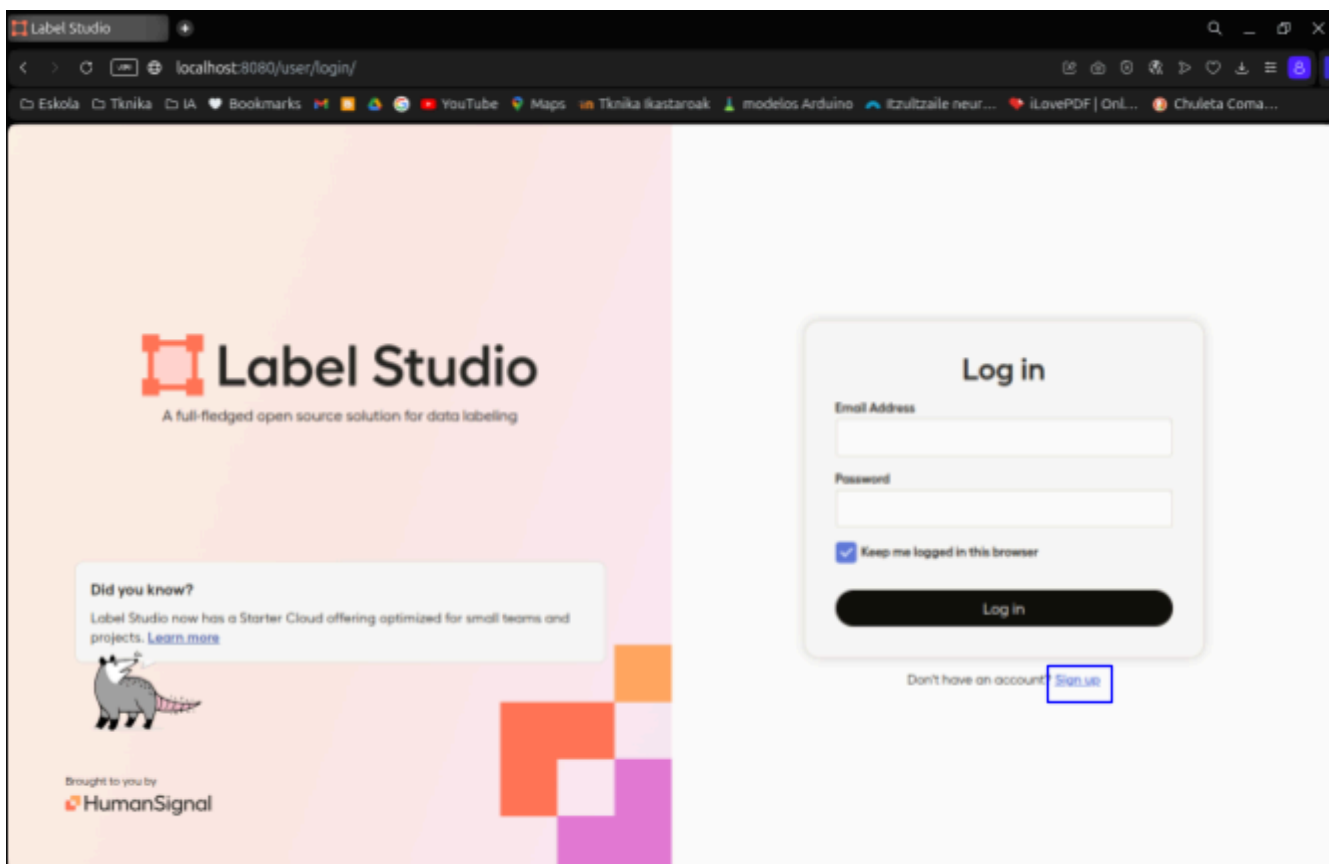
Parameter	Explanation
<code>-p 8080:8080</code>	Web interface on port 8080
<code>-in ...</code>	Data storage volume
<code>-it</code>	Interactive terminal

2. Use Label Studio

As mentioned in the previous point, to get started with Label Studio, you need to have a docker container running. In a browser, we will go to this address:

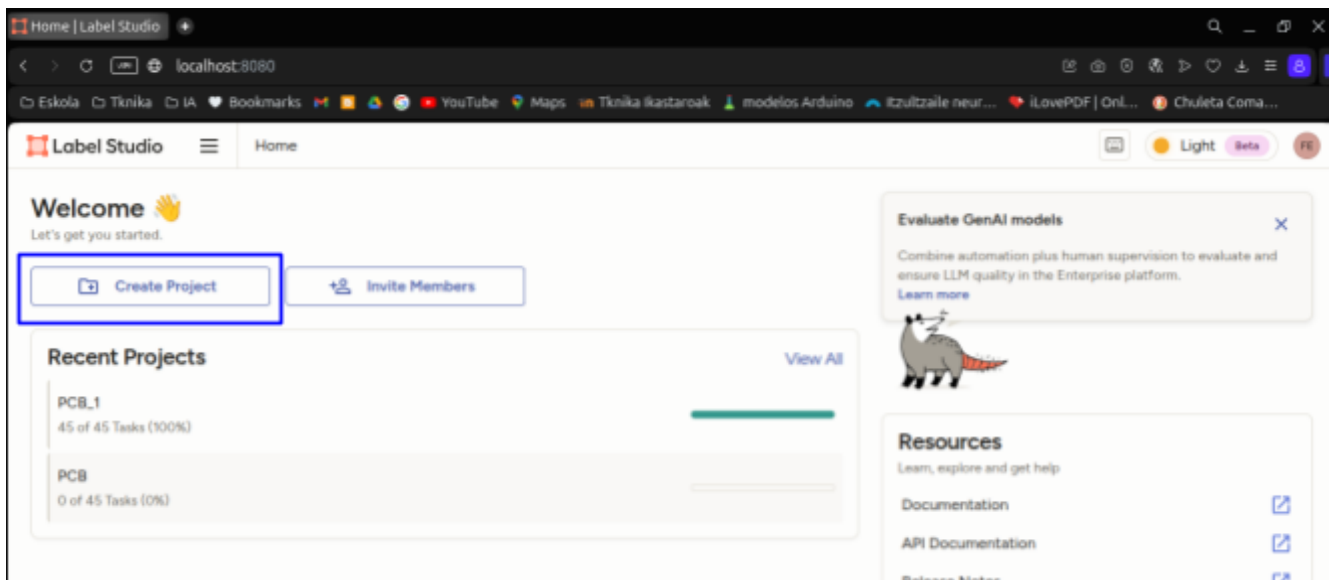
```
http://localhost:8080/user/login/
```

The web page that will open will be:



When we log in for the first time, we will have to create a user, it is advisable to save the user to always have control over our projects. If we lose the user, we can create a new one, but we would lose the projects we have made.

To create our project we will click the CREATE PROJECT button:



The following screen will appear and we must fill in the three fields there with some minimum information.

Project Name

The screenshot shows the 'Create Project' dialog box. At the top, there are three tabs: 'Project Name', 'Data Import', and 'Labeling Setup'. The 'Project Name' tab is active. Below the tabs, there are three main sections: 'Project Name' with a text input field containing 'New Project #3', 'Description' with a text area containing 'Optional description of your project', and 'Workspace' with a dropdown menu showing 'Enterprise'. Below the 'Workspace' section, there's a note: 'Simplify project management by organizing projects into workspaces. [Learn more](#)'. At the bottom, there's a 'Did you know?' section with a cartoon dinosaur and the text: 'Try Label Studio Starter Cloud, optimized for small teams and projects. [Learn more](#)'. At the top right of the dialog, there are 'Cancel' and 'Save' buttons.

The facts would suffice, no explanation is necessary, and the last Workspace section is the part for companies.

Data Import

Create Project

Project Name **Data Import** Labeling Setup

Cancel Save

Dataset URL Add URL or Upload Files

**Drag & drop files here
or click to browse**

Images	bmp, gif, jpg, jpeg, png, svg, webp
Audio	wav, mp3, flac, m4a, ogg
Video	mp4, webm
HTML / HyperText	html, htm, xml
Text	txt
Structured data	csv, tsv, json
PDF	pdf

Important:

- We recommend [Cloud Storage](#) over direct uploads due to [upload limitations](#).
- For PDFs, use [multi-image labeling](#). JSONL or Parquet (Enterprise only) files require cloud storage.
- Check the documentation to [import preannotated data](#).

There we will upload all the photos we will tag. And finally, and perhaps most importantly, the third page:

Labelling Setup

Create Project

Project Name Data Import **Labeling Setup**

Cancel Save

Computer Vision >

- Natural Language Processing >
- Audio/Speech Processing >
- Conversational AI Chat >
- Ranking & Scoring >
- Structured Data Parsing >
- Time Series >

Semantic Segmentation with Polygons

Semantic Segmentation with Masks

Object Detection with Bounding Boxes

OCR Labeling for PDFs

Select task to connect

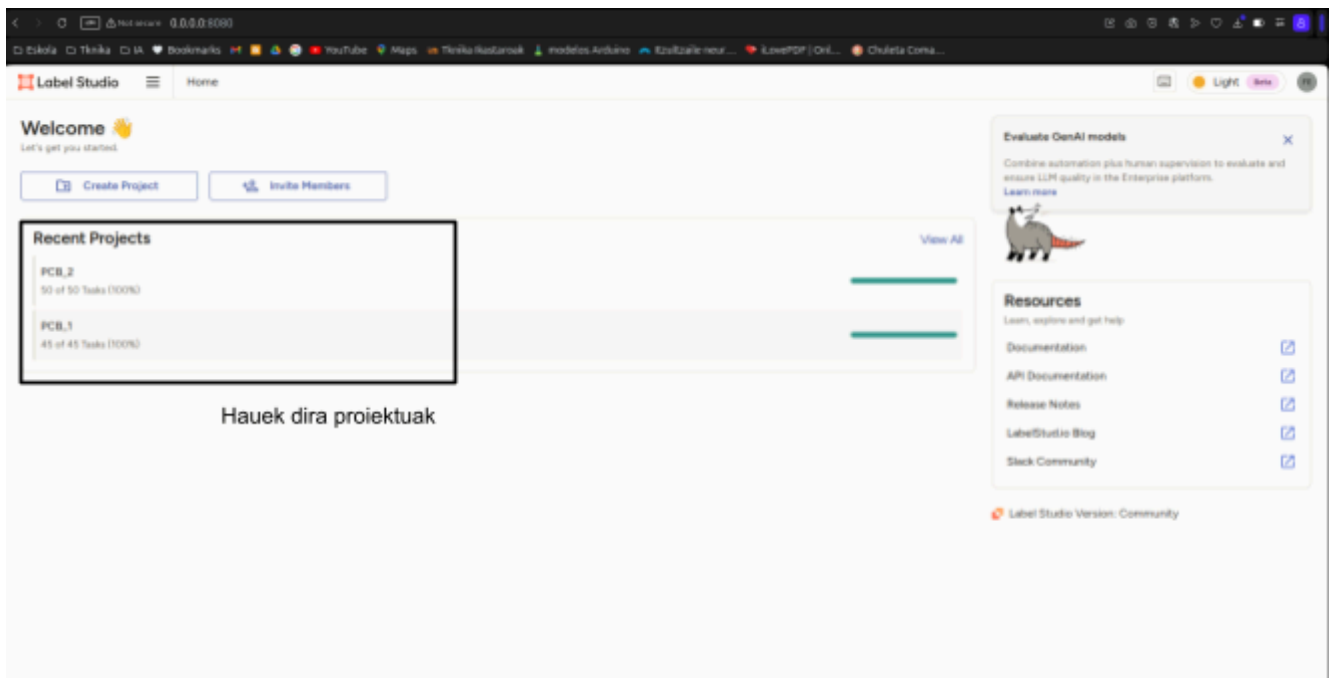
Type: **Object Detection** **Enterprise**

Opossum Cuteness: A comparative analysis

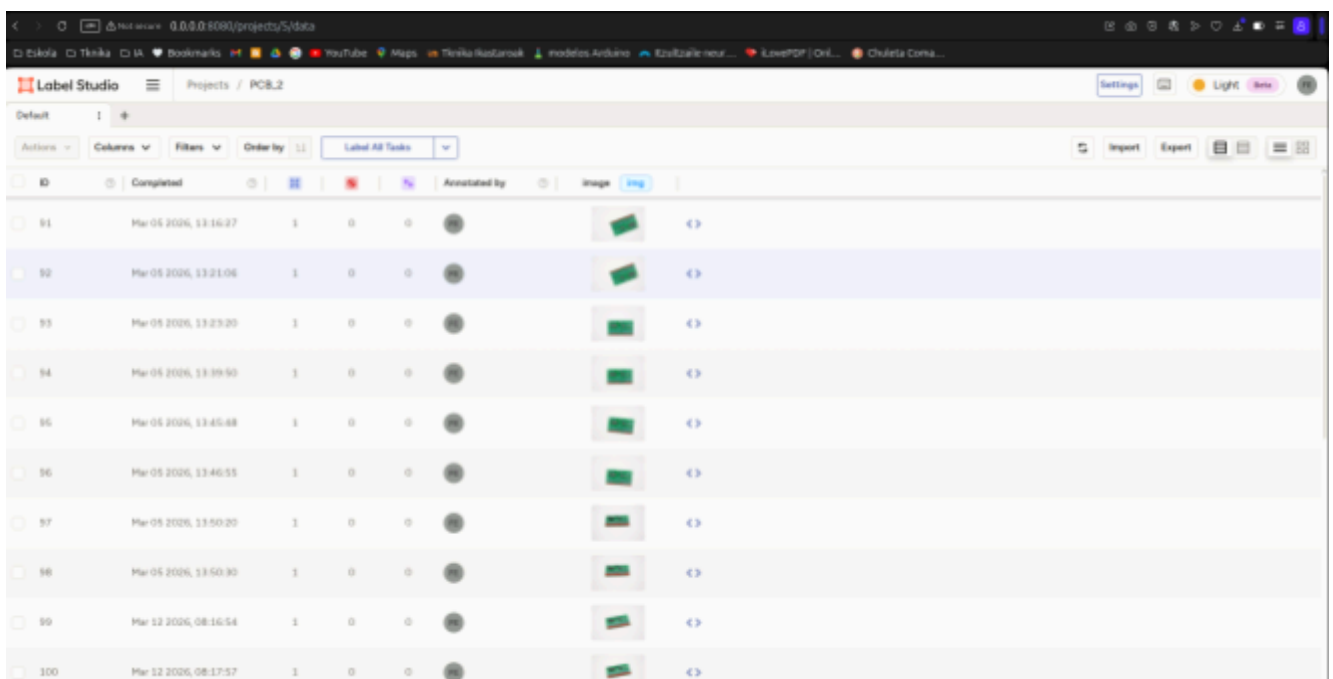
Enterprise

Here it is very important what type of project we are going to do the labeling for. In our case we will select COMPUTER VISION on the left and OBJECT DETECTION WITH BOUNDING BOXES from the options that appear.

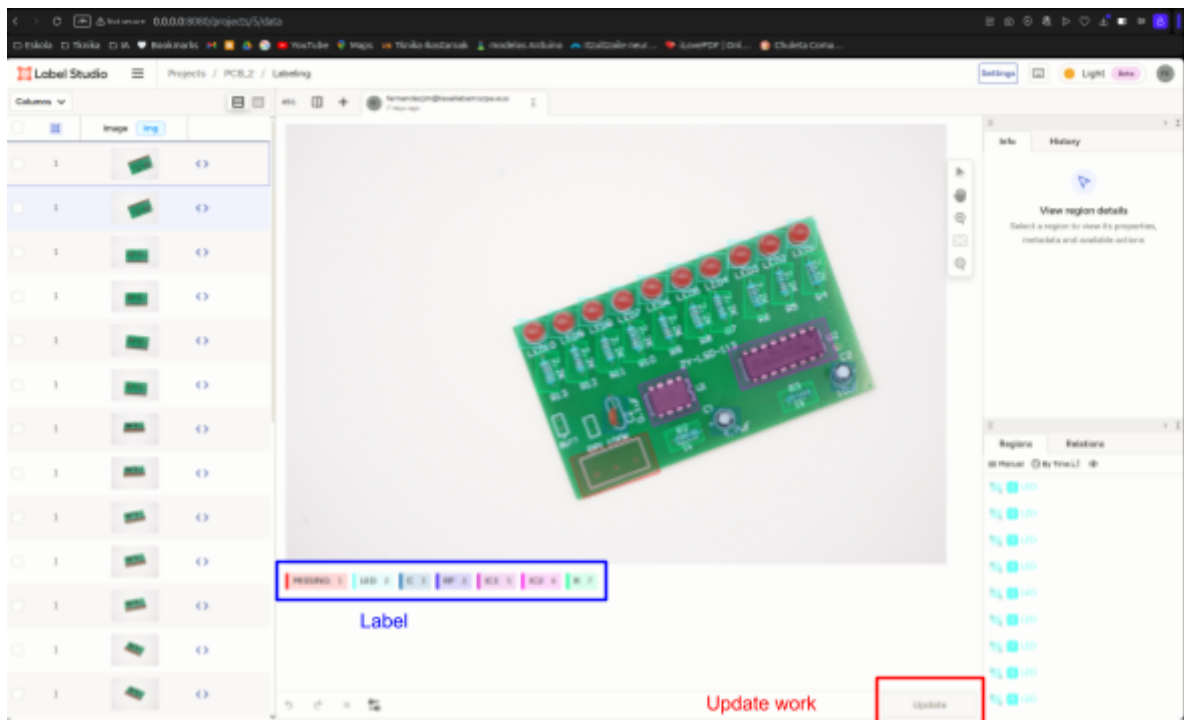
We have our project up and running and now we will label the photos we have uploaded. To do this, go to the Project page and this will appear:



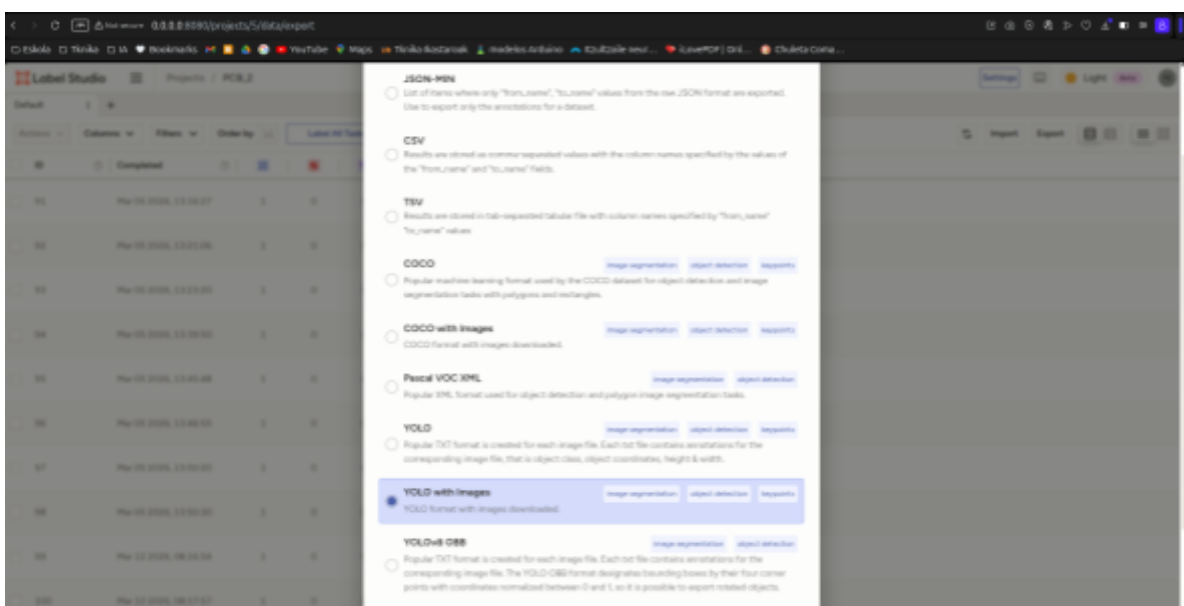
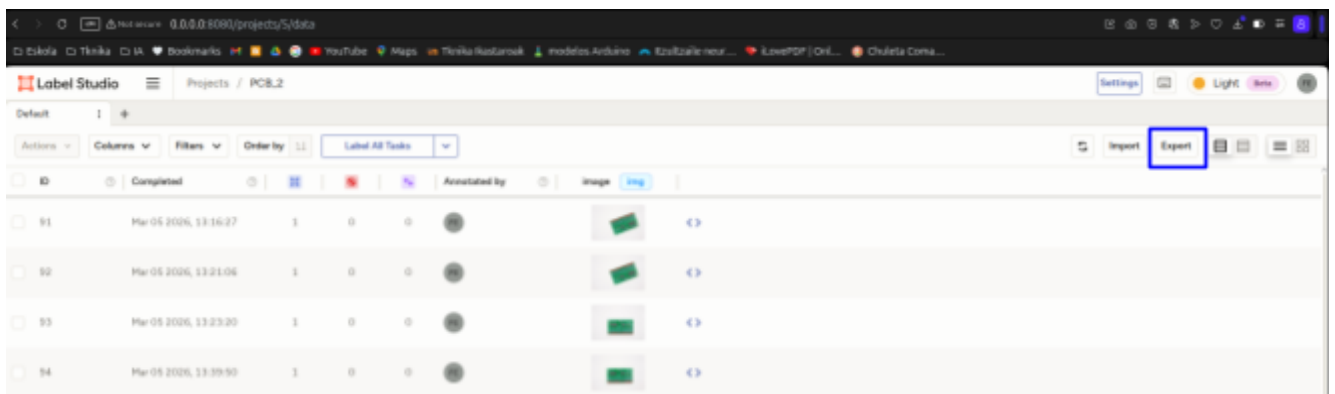
We will access the created project by clicking on it and we will see a screen like this:



And finally, to add labels, click on each photo and the labels created will appear on the screen like this. Choose a label and make a box where you want to put it and put the first label. Put all the labels you want on the photo and click on the button that appears on the back. When you do it for the first time **Submit** button will appear and when you edit or move a label **Update** will put it on the button.



The last step will be to export the work done. In our case we use YOLO and choose what we want to export. **YOLO with images** it will be.



3. Take pictures

After doing some research, we took the photos we needed for the training. On the one hand, a light box was purchased, in our case, and we chose this light box because the photos we needed to take were of PCB boards.



This PULUZ lighting box measures 50x50x50 cm, large enough to hold PCB boards. It comes with its own lighting system and integrates a lighting control system. It has a hole in the top that allows you to take vertical photos. It also has a photo diffuser and different colored backgrounds. The color of the light can be changed, between cool and warm light. This allows for photos with different colors, intensities and backgrounds, thus increasing the effectiveness of your training.

You don't need a very good camera to take a photo, but the resolution of the photo should be as good as possible. The camera I used is:



The resolution of this camera is 24 Mpixels and the size of the images it produces is 6000 x 4000 at its largest size. It can take RAW photos but for training photos it is better to take them directly **.jpg**. We took them in the format, without editing the photo.

At first we thought about taking the photos with a fixed camera, but the photos we took and the position of the plate changed, and they came out very homogeneous. So in the end we took the camera in our hands and took the photos at different angles, positions and lighting conditions. [Here is a collection of those photos.](#)